

(worn) on the head of a person. At times they are also called as ‘helmet mounted displays’. HMDs are of two types out of which one cannot be used for AR.

1 **HMDs that display only CGI** - Most HMDs that you can find in the common market (though you will not be able to find them easily) are of this type and display only computer generated images (CGI). These HMDs are the ones which cannot be used for AR. The main requirement of AR is to augment the reality. Since these HMDs do not feature a real-world view capability in the first place, the question of augmenting the view gets out of scope. These HMDs are usually used in virtual reality setups.

2 **HMDs that allow superimposition of CGI with real world view** - These displays are capable of displaying real world imagery on which superimposition of additional data can be done. These once again are of two types:

a **Video see through:** These HMDs take in the real view via a camera and mix the CGI information onto it.

b **Optical see through:**

These HMDs allow the real world to be viewed through a semi-reflective mirror and project the CGI data on to the mirror using a projector device. The mirror being half reflective only, creates the superimposition effects. Yet another type of optical see through HMDs are the ones which use OLED displays which act as both viewing glass as well as displays.



One of the various types of HMDs

HMDs are one of the most useful type of displays although they are not the most common ones. The usage of HMDs are many.